Page 1: Operating System Fundamentals

Document date: 23 September 2026

An operating system manages hardware and provides services to applications. It controls the CPU, memory, storage, devices, files, security, and user interaction.

Key points

• Understand the definition and purpose of the concept.

• Identify the major components, algorithms, or conditions involved.

• Compare related approaches and understand their practical trade-offs.

• Use this section to test document retrieval, chunking, embeddings, and question answering in the IntelliLearn RAG pipeline.

Page 2: Processes and Threads

Document date: 23 September 2026

A process is a program in execution. Process state, PCB, context switching, process creation, and termination are fundamental OS concepts. Threads provide smaller units of execution within a process and can share process resources.

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Page 3: CPU Scheduling

Document date: 23 September 2026

CPU scheduling selects a ready process for execution. FCFS, SJF, SRTF, Priority Scheduling, and Round Robin have different trade-offs involving waiting time, turnaround time, response time, throughput, and fairness.

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Page 4: Process Synchronization

Document date: 23 September 2026

Concurrent processes can create race conditions when they access shared data. Critical sections, mutexes, semaphores, monitors, and synchronization protocols help ensure correct execution.

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Page 5: Deadlocks

Document date: 23 September 2026

A deadlock occurs when processes wait indefinitely for resources held by other processes. Mutual exclusion, hold and wait, no preemption, and circular wait are the four necessary conditions.

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Page 6: Memory Management

Document date: 23 September 2026

The operating system allocates and tracks memory. Paging divides logical memory into pages and physical memory into frames. Page tables translate logical addresses into physical addresses.

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Page 7: Virtual Memory

Document date: 23 September 2026

Virtual memory lets processes use logical address spaces larger than physical RAM. Demand paging loads pages when needed, while page faults trigger retrieval of missing pages from secondary storage.

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Page 8: Page Replacement

Document date: 23 September 2026

When no free frame exists, the OS selects a page for replacement. FIFO, Optimal, and LRU are common algorithms. Their behavior can be compared using page-fault counts.

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Page 9: File Systems

Document date: 23 September 2026

File systems organize persistent data using files and directories. They manage metadata, permissions, allocation, free space, and recovery. Common allocation approaches include contiguous, linked, and indexed allocation.

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Page 10: I/O and Storage

Document date: 23 September 2026

I/O management uses device drivers, interrupts, buffering, caching, and DMA. Disk scheduling algorithms such as FCFS, SSTF, SCAN, and C-SCAN manage pending storage requests.

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Page 11: Security and Protection

Document date: 23 September 2026

Authentication establishes identity while authorization controls access. Protection mechanisms isolate processes and resources, and least privilege limits permissions to what is necessary.

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Page 12: System Calls

Document date: 23 September 2026

System calls provide the interface between application programs and kernel services. Common categories include process control, file management, device management, communication, and protection.

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Page 13: Networking in Operating Systems

Document date: 23 September 2026

Operating systems provide networking through sockets and protocol stacks. TCP provides reliable ordered delivery, while UDP offers connectionless communication with lower overhead.

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Page 14: Performance and Monitoring

Document date: 23 September 2026

Useful OS metrics include CPU utilization, throughput, response time, waiting time, memory utilization, I/O activity, and page-fault rate. Monitoring helps identify system bottlenecks.

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Page 15: Review and RAG Questions

Document date: 23 September 2026

Review questions can test definitions, comparisons, calculations, and explanations. For RAG testing, questions should be answerable from retrieved document chunks and should preserve page or section metadata.

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